Characterization of a Plastid Genome

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BIO 300-B

Genome characterization summary

Item	Result
Genus and species	Glycine soja (wild soybean), isolate HAAS216
Family	Fabaceae
Database / accession-version	NCBI Nucleotide / GenBank, PX591265.1
Record title	Glycine soja isolate HAAS216 chloroplast, complete genome
Genome length	152,222 bp
Topology	Circular, as reported in the GenBank LOCUS line
Galaxy sequence records	1
GC content	A = 49,270; T = 49,107; C = 26,485; G = 27,360; N = 0
Base counts	128 gene; 83 CDS; 37 tRNA; 8 rRNA features
Annotation feature rows	109 total: 76 protein-coding names, 29 tRNA names, 4 rRNA names
Distinct gene names	109 total: 76 protein-coding names, 29 tRNA names, 4 rRNA names
Genome regions	Coordinate-based estimates: LSC ≈83,179 bp; SSC ≈17,873 bp; IRa and IRb ≈25,585 bp each. These are estimates, not explicit region annotations in this GenBank record.
Intron / RNA processing	11 protein-coding genes have joined CDS locations consistent with introns; rps12 is trans-spliced. Eight tRNA feature rows also have joined locations.
Duplications / pseudogenes	19 gene names occur in multiple gene features, largely associated with IRs; no /pseudo or /pseudogene qualifier

1. Full scientific name, family, accession, source, and genome size

The selected organism is Glycine soja (wild soybean), isolate HAAS216, in the family Fabaceae. The complete chloroplast genome record is available from NCBI Nucleotide/GenBank as accession PX591265.1. Its length is 152,222 base pairs and the LOCUS line reports circular topology. The record title is “The complete chloroplast genome of wild soybean (Glycine soja) isolate HAAS216,” attributed to Wang and Bi. The record identifies the submission as unpublished/direct submission rather than a journal article. NCBI record: https://www.ncbi.nlm.nih.gov/nuccore/PX591265.1

2. Evidence that the sequence is a complete plastid genome

The NCBI definition explicitly identifies this record as a “chloroplast, complete genome.” The source is chloroplast DNA from Glycine soja, and the GenBank LOCUS line reports a circular 152,222-bp DNA molecule. The sequence is represented as one record in Galaxy, and its annotation includes multiple expected plastid gene groups: photosystem genes (psa and psb), ATP synthase genes (atp), cytochrome b6f genes (pet), rbcL, RNA polymerase genes (rpo), ribosomal protein genes (rpl and rps), rRNA genes (rrn), and tRNA genes (trn). This combination supports that the record is a complete plastome rather than a single barcode gene or a short fragment.

3. Overall genome organization and LSC/SSC/IR regions

The genome is consistent with the common quadripartite organization found in many land-plant chloroplast genomes: a large single-copy (LSC) region, an inverted repeat (IRb), a small single-copy (SSC) region, and a second inverted repeat (IRa). Based on the duplicated-gene boundary coordinates in the supplied GenBank annotation, approximate sizes are LSC ≈83,179 bp, IRb ≈25,585 bp, SSC ≈17,873 bp, and IRa ≈25,585 bp. These estimates sum to 152,222 bp. The GenBank record does not explicitly annotate named LSC, SSC, IRa, or IRb region features, so the values should be treated as coordinate-based estimates and verified with a plastome map or IR-detection tool if exact boundaries are required.

4. Annotated gene content, gene counts, pseudogenes, and IR duplication

The GenBank feature table contains 128 gene feature rows, 83 CDS feature rows, 37 tRNA feature rows, and 8 rRNA feature rows. Counting distinct gene names instead gives 109: 76 protein-coding gene names, 29 tRNA gene names, and 4 rRNA gene names. These numbers differ because genes may occur in duplicated copies, and a feature row is not the same as a unique gene type. The four rRNA types are rrn16S, rrn23S, rrn4.5S, and rrn5S; each appears twice in the annotation.

Gene count item	Count / result	Interpretation
Gene feature rows	128	Rows of type gene in the GenBank feature table.
Unique gene names	109	Distinct names across gene features.
Protein-coding gene names	76	Distinct gene names among CDS features.
CDS feature rows	83	Includes duplicated copies and split/joined coding locations.
tRNA gene names / features	29 names / 37 features	Some tRNA types are duplicated in the genome.
rRNA gene names / features	4 names / 8 features	Four rRNA types, each annotated twice.
Intron-containing protein-coding genes	11 likely intron-containing genes	ycf3, rpoC1, atpF, rps16, clpP, petB, petD, rpl16, rpl2, ndhB, ndhA.
Trans-spliced gene	rps12	Exons are separated in the genome and joined during RNA processing; this is not simply a conventional intron example.
Repeated gene names	19	Repeated gene names include IRassociated copies; not all repeats indicate a new gene type.
Pseudogene qualifiers	0 found	No /pseudo or /pseudogene qualifier was found in the supplied annotation; this does not prove all genes are functional.

The two IRs are inverted copies of the same sequence and commonly contain rRNA genes and selected tRNA/protein-coding genes. Repeated names in this record include rrn16S, rrn23S, rrn4.5S, rrn5S, rpl2, rpl23, ndhB, rps7, ycf2, and several tRNA genes. A partial ycf1 copy is annotated near an IR/SSC boundary. Duplicate feature rows should not be counted as distinct gene types.

5. At least eight protein-coding genes and their functions

Gene	Functional group	Biological function
psaA	Photosystem I	Encodes a core Photosystem I reaction-center protein involved in light-driven electron transfer.
psbA	Photosystem II	Encodes the D1 reaction-center protein, central to Photosystem II electron transfer.
atpB	ATP synthase	Encodes the beta subunit of ATP synthase, which helps produce ATP using the proton gradient.
petB	Cytochrome b6f	Encodes cytochrome b6, part of the complex linking photosynthetic electron-transfer steps.
rbcL	Carbon fixation	Encodes the large subunit of Rubisco, which fixes CO_2 in the Calvin cycle.
rpoB	Transcription	Encodes the beta subunit of the plastid-encoded RNA polymerase.
rpl2	Ribosome	Encodes a large-subunit ribosomal protein required for plastid protein synthesis.
ndhA	NDH-like complex	Encodes a subunit of the plastid NADH dehydrogenase-like complex involved in photosynthetic electron transport.
matK	RNA processing	Encodes maturase K, associated with splicing certain group II introns.
clpP	Protein quality control	Encodes the proteolytic subunit of the Clp protease complex, involved in protein turnover.

6. and RNA-processing features

Four distinct rRNA gene types are annotated: rrn16S, rrn23S, rrn4.5S, and rrn5S. Each has two feature entries, consistent with copies in the inverted-repeat regions. Examples of tRNA genes include trnH-GUG (histidine), trnM-CAU (methionine), trnF-GAA (phenylalanine), trnI-GAU (isoleucine), trnA-UGC (alanine), and trnV-UAC (valine). There are 37 tRNA feature rows representing 29 distinct tRNA gene names.

Examples of protein-coding genes with joined CDS locations consistent with introns are ycf3, rpoC1, atpF, rps16, clpP, petB, petD, rpl16, rpl2, ndhB, and ndhA. The rps12 coding sequence is trans-spliced: its exons occur at separate genome locations and are joined during RNA processing. Eight tRNA feature rows also have joined locations, which is a reminder that RNA processing can involve tRNA introns as well.

7. Pseudogenes, gene losses, duplications, rearrangements, and unusual features

No /pseudo or /pseudogene qualifier was found in the GenBank annotation reviewed. Gene duplication is evident, especially in the IR regions. Examples of repeated gene names include rrn16S, rrn23S, rrn4.5S, rrn5S, ndhB, rpl2, rpl23, ycf2, and multiple tRNA genes. A partial ycf1 copy occurs near an IR/SSC boundary, and rps12 is trans-spliced. The record by itself does not establish whether genes were lost or whether rearrangements occurred relative to an ancestor; those claims would require a comparative analysis with related species.

8. GC content and two additional observations

Galaxy sequence statistics report 152,222 bp, one sequence record, and 35.37% GC. The base counts are A = 49,270, T = 49,107, C = 26,485, G = 27,360, and N = 0. Thus, the sequence is AT-rich: A and T together make up approximately 64.63% of the bases.

1. The genome is represented by one sequence record with no ambiguous N bases, consistent with a single complete plastome sequence in the FASTA file.
2. The annotation shows duplicated IR-associated gene copies and multiple joined coding locations; examples include ycf3 and rpoC1, while rps12 is trans-spliced. The sequence N50 and N90 are both 152,222 bp because the input contains one sequence record.

9. Plastid versus mitochondrial genomes: similarities and differences

Similarities

- Both are organellar genomes located outside the nucleus, in plastids/chloroplasts and mitochondria, respectively.
- Both encode a subset of proteins and RNAs needed for organelle function.
- Both are usually present in multiple DNA copies per cell or organelle, although copy number varies with tissue and development.
- Both have bacterial evolutionary origins through endosymbiosis.
- Both have transferred many ancestral genes to the nuclear genome during evolution.
- Both are useful in evolutionary, phylogenetic, population, and lineage studies.

Differences

- Location and main role: plastids support photosynthesis and other plastid functions; mitochondria carry out cellular respiration and oxidative phosphorylation.
- Gene content: plastomes typically encode photosynthesis-related genes such as psa, psb, pet, and rbcL; mitochondrial genomes typically encode respiratory-chain components and mitochondrial translation machinery.
- Organization: many land-plant plastomes are compact and have an LSC–IR–SSC–IR arrangement; plant mitochondrial genomes can be structurally dynamic and may exist as recombining/branched or linear forms rather than a single simple circle.
- Relative size: land-plant plastomes are often around 120–160 kb, while plant mitochondrial genome sizes vary widely and are often larger.
- Gene order and structural change: plastome gene order is relatively conserved in many land plants; plant mitochondrial genomes often show frequent repeat-mediated recombination and rearrangements.
- Mutation patterns: plant mitochondrial nucleotide substitution rates are often slow despite structural dynamism; plastid substitution patterns differ and vary among lineages and regions.
- Inheritance: both are often uniparentally inherited in flowering plants, commonly maternally, but exceptions occur and inheritance should be checked for the lineage.
- Research emphasis: plastid genomes are especially useful for photosynthetic evolution and plant lineage tracking; mitochondrial genomes are useful for respiration-related evolution, mitochondrial inheritance, and cytoplasmic male sterility.

Feature	Plastid genome	Mitochondrial genome
Cellular location	Plastids/chloroplasts.	Mitochondria.
Main biological functions	Photosynthesis, carbon fixation, and plastid gene expression.	Respiration, oxidative phosphorylation, and energy metabolism.
Typical organization	Often compact; many land plants have LSC, SSC, and two IRs.	Highly variable in plants; may be structurally dynamic and recombining.
Relative size	Often about 120–160 kb in land plants.	Varies widely; often larger than plastomes in plants.

Gene content	Photosynthesis genes, RNA polymerase, ribosomal proteins, tRNAs, rRNAs.	Respiratory-chain genes, mitochondrial translation genes, tRNAs/rRNAs, other organelle genes.
Copy number	Multiple copies; varies by tissue, cell, and developmental stage.	Multiple copies; varies by tissue, cell, and developmental stage.
Inheritance	Often maternal in angiosperms, with exceptions.	Often maternal in flowering plants, with exceptions.
Recombination / structural change	Gene order often relatively conserved; IRs are a common structural feature.	Repeat-mediated recombination and rearrangements are common in many
		plant lineages.
Mutation / substitution pattern	Rates vary; noncoding regions can be informative for phylogeny.	Nucleotide substitution often slow in plants despite structural changes; varies by lineage.
Common research applications	Plant phylogeny, barcoding, photosynthesis evolution, crop lineage tracking.	Respiration evolution, cytoplasmic male sterility, mitochondrial inheritance and evolution.

10. Practical value and limitations of plastid genomes

Advantages compared with the nuclear genome

- The plastome is relatively small, making sequencing, storage, assembly, and analysis more manageable.
- Plastid DNA can occur at high copy number in plant tissues, which can improve recovery from some samples. Many plastomes have conserved gene order, supporting alignment and comparison among related taxa.
- A complete plastome provides many loci in one linked sequence rather than relying on a single barcode marker.
- Plastid data can support plant identification, phylogeny, phylogeography, lineage tracing, and crop domestication studies.
- When inheritance is known, plastid genomes can help trace maternal or other cytoplasmic lineages.
- Plastid genes can be used to study photosynthesis-related variation and plastid evolution.

Limitations

- A plastome represents one organellar lineage and often behaves as one linked locus, so it may not reflect the full species history.
- It generally samples less genome-wide variation than the nuclear genome and may not resolve very recent divergences.
- Inheritance is often uniparental but varies among taxa and should not be assumed without evidence.
- It does not capture most nuclear traits, genome-wide recombination, or the full genetic basis of complex traits.
- Contamination, assembly artifacts, and gene transfer between the nucleus and organelles can complicate interpretation.
- Plastid data alone cannot resolve questions requiring biparental ancestry, genome-wide admixture, or nuclear sex-chromosome variation.

Research question suited to plastid data: Which chloroplast lineages are shared among wild soybean populations collected from different geographic regions?

Research question better suited to nuclear genomic data: Which genome-wide nuclear variants and alleles contribute to drought tolerance or domestication traits in soybean?

Data source and references

NCBI Nucleotide / GenBank. Glycine soja isolate HAAS216 chloroplast, complete genome. Accession PX591265.1. https://www.ncbi.nlm.nih.gov/nuccore/PX591265.1

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